R16

Code No: 135BY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
COMPUTER GRAPHICS
(Common to CE, CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Discuss in detail Bressman's Algorithm.
b) Find the coordinates of pixels that lie on line segment having end points (5,8) and (2,4). [7+8]

2.a) Write a note on various kinds of video display devices.
b) Write boundary filling algorithm. [8+7]

3.a) Discuss the stages involved in 2D view transformation pipeline.
b) Write polygon clipping algorithm. [8+7]

4.a) Explain Cohen Sutherland algorithm.
b) Explain composite transformation for translation, scaling and rotation. [7+8]

5.a) Explain 3D viewing in detail.
b) Derive the transformation matrix for Bezier curve. [7+8]

6. Derive the transformation matrix for rotation about an arbitrary axis. [15]

7.a) Explain classification of visibility surface detection algorithms.
b) Explain about constant-intensity shading algorithm. [8+7]

8.a) Explain about the computer animation languages.
b) Explain about the key frame system. [7+8]

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